

Data and Computing Resource Management Plan

1. Introduction

The goal of this Data and Computing Resources Management Plan is to provide a systematic, centralized, and expertly supported approach to meeting the data management and cyberinfrastructure needs of the Odor2Action network. For a project of this size, complexity, and interdisciplinarity, significant infrastructure and expertise are required to enable robust large-scale centralized data storage, seamless access to data and computational resources, and open science practices like public access to well-curated data, code, and software in established open repositories. This proposed plan adds value by providing a means for secure and efficient transfer of large datasets between theoreticians and experimentalists. As noted in the recent NSF Dear Colleague Letter¹⁶⁶ NSF 19-069 “Effective Practices for Data,” open science practices have myriad benefits for the scientific and broader community; however, making data openly accessible and actually useful via expert curation requires personnel effort and technical resources. This plan leverages existing and past NSF investments to ensure that data are easily and safely stored in centralized infrastructure with simple and fast access to data and analysis tools by the large and geographically distributed project team. In addition, the plan enables openly available and reusable data in a sustainable way over the long term, according to emerging best practices. These practices include the use of persistent identifiers such as DataCite Digital Object Identifiers (DOIs), a machine-readable data management plan as described in NSF document 19-069 referenced above, and adherence to the broader FAIR data principles and emerging software citation principles.^{215,258} This plan describes how the variety and volume of data produced will be supported with appropriate expertise and resources for the entire lifecycle from data storage and analysis to open data sharing and long-term preservation.

2. Cyberinfrastructure Team

Digital data and computing resources will be managed by the Center for Research Data and Digital Scholarship (CRDDS) at the University of Colorado Boulder. CRDDS brings together national experts in cyberinfrastructure and data management to support research data, computation, and analysis across disciplines in accordance with open science principles.

Senior Personnel Thomas Hauser: Hauser is CRDDS executive director and director of research computing at CU Boulder. He will oversee cyberinfrastructure and guide the use of the computing resources for the project. Hauser is PI on the NSF Hybrid Cloud award, the NSF Cyberteam award and is overseeing the operations and support of the central computing and data resources for the university. His group operates an advanced computational and data-intensive infrastructure consisting of supercomputers, high-throughput computing and data resources connected through a Science DMZ to other regional and national resources. CRDDS offers a large number of training programs for researchers and support for data-intensive research, and Research Computing provides user support for the compute and storage systems.

Senior Personnel Andrew Johnson: Johnson is CRDDS director of research data management and associate professor at CU Boulder. He will oversee data management, data archiving, and data sharing/publishing for the project. Johnson’s team supports data management best practices according to national and international standards and runs the CU Boulder repository, CU Scholar, which provides infrastructure for data publishing and archiving in accordance with open science principles. Johnson has a strong record of publications and grants, including from the Institute of Museum and Library Services, in the area of data management support.

3. Cyberinfrastructure System Description

Hauser and Johnson will develop and support a system of interconnected resources to ensure that data move as easy as possible from instruments and other data collection sources to a centralized storage system (PetaLibrary), where it can be accessed by the project team for analysis and computation (Fig. 19). The data on the PetaLibrary can be accessed by researchers through

web-based interfaces like Jupyter notebooks¹³⁰ or the OpenOnDemand portal.¹⁰⁶ The data analysis will be performed on several computing resources. The project has access to the five compute nodes budgeted for purchase in the Blanca cluster. These resources are dedicated to the project, but other idle compute cycles of the system can also be utilized by this project through preemptible queues. Additionally, the project will have access to the RMACC Summit supercomputer (funded by an NSF MRI award) for computing needs that go beyond the nodes of the Blanca cluster. Hybrid cloud computing resources will also be available to the project through the NSF CC*-funded hybrid cloud infrastructure that will be operational in the spring of 2020. In addition, data stored in the PetaLibrary can be shared easily with collaborators or with anyone who requests access, even to large data sets, using the Globus data transfer service. Data sets stored in the PetaLibrary that are associated with publications or of potential reuse by the community can be more easily curated for open science publication and long-term archiving in repositories supported by the team (CU Scholar) as well as external repositories (Zenodo, OdorMapDB).

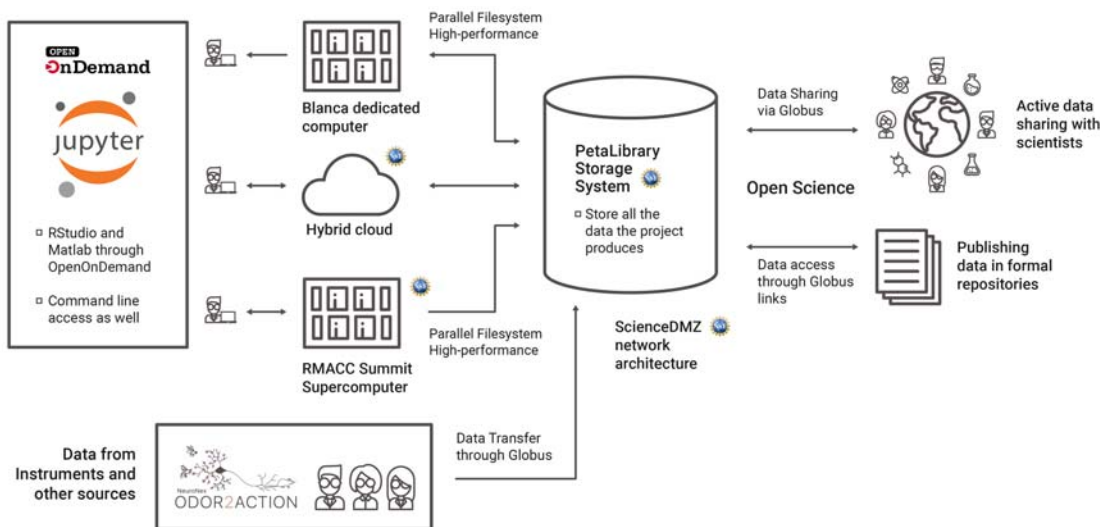


Figure 19: Schematic of the Odor2Action cyberinfrastructure system for data sharing, computing, and open science. NSF Logos denote leveraging of previous NSF cyberinfrastructure investments.

Types of data: All behavioral, imaging, and *in vivo* neurophysiological data will be organized by trial/session/animal. Behavioral data includes video data and automated measurements characterizing the behavior such as velocity, position, and sensor/head motion. Imaging data trials consist of image stacks collected at 30 frames per second and digitized at 16-bit precision. Trials also include time series (50 samples/s) of miniPID olfactometer odor concentrations, licking and sniffing behavior (for mice), and metadata (odor, concentration, plume ID, other experimental information). *In vitro* and other physiological data will include time series of voltage and/or current as well as metadata on cell location and morphology. Other datasets will consist of odor plume measurements, robot trajectory data, simulation results, Illumina sequence data, and video data. Code for simulation and analysis of the experimental data will be in Python, R, or MatLab. Data is expected to total approximately 150 terabytes in size, necessitating the use of a large-scale storage solution and specialized tools and workflows for data sharing, publishing, and archiving as described below.

Data standards: Voltage data, images, and other large datasets will be kept in raw format along with an easily readable file that contains information about the type of data, the stimuli, and related information about the experiment. The project will maintain a common format for each kind of experimental data, and establish a data processing pipeline that, when possible, is shared across labs in the different IRGs. The pipeline will yield data that is stored in an open-source file structure to facilitate sharing (e.g., HDF5). Documentation for each kind of file will include example scripts

(in Python or MatLab) that access the data. Johnson will assist with metadata creation consistent with international standards (e.g., DataCite metadata schema) to promote discovery, sharing, and reuse of data. DataCite DOIs will be registered by the CU Boulder CRDDS for all data that is published via the CU Scholar repository in order to provide persistent and citable links to the data and metadata. Publishing data with DataCite DOIs supports key open science practices, including the FAIR data principles.

Plans for data storage: The CU Boulder “PetaLibrary” service provides a stable, scalable, and cost-effective solution for the storage and archiving of research data. The PetaLibrary enables researchers to build, store, share, and merge large and growing data collections, and is available at a subsidized cost to researchers affiliated with the University of Colorado Boulder. Storage is provided in two primary classes: “active” storage for data that is accessed frequently or is still undergoing analysis, and “archive” storage for data that is more static but needs to be stored securely for a longer-term. Active storage is accessible from all Research Computing compute resources through the BeeGFS parallel file system. Archive storage resides on a cost- and energy-efficient hierarchical system backed by LTO tape. Both classes of storage can be accessed by any authorized user from any Internet-connected computer using several common data-transfer protocols. The PetaLibrary infrastructure is housed in a controlled-environment data center protected by an uninterruptible power supply (UPS) system. The PetaLibrary is supported by a high-speed local network and dedicated large-bandwidth connections to national research networks through an initially NSF-funded ScienceDMZ. By using the PetaLibrary as a centralized storage solution, the project team will be able to collaboratively analyze the data of the project, share it with other collaborators outside of the core team, and publish and archive data more easily via the Globus connection between the PetaLibrary and the CU Scholar repository. Non-digital data and other physical resources will be stored and managed by the individual IRGs.

Active access to the data and compute environment: All Odor2Action collaborators will have direct access to all compute and data resources funded through this project as well as general available NSF-funded infrastructure via affiliate accounts. The project team has funded compute nodes in the Blanca cluster for exclusive access as well as a node in the hybrid cloud. All the data on the PetaLibrary can be shared using Globus. The project team can share the data with anyone without providing accounts on the CU Boulder system. Globus even allows the project team to create and manage Globus groups for access to the shared data. Globus also provides a mechanism for public access to large published data sets on the PetaLibrary via landing pages in the CU Scholar repository. The project team will have Linux command line access to the compute nodes purchased for this project. These compute nodes will be integrated into Research Computing condo computing service “Blanca”. The project team has immediate access to its nodes when needed. Additionally, they can utilize free nodes of the aggregate Blanca cluster. Every Research Computing user has access to the NSF funded RMACC Summit supercomputer for parallel computation in a shared environment. Home, project, and the PetaLibrary storage is accessible on all computing resources. Research Computing is providing a JupyterHub and OpenOnDemand environment to support access to the computing and data resources through Jupyter notebooks and an Rstudio environment, along with MATLAB. CRDDS and Research Computing will provide training in all aspects of cyberinfrastructure use, and will support development of data and analytics workflows.

Plans for archiving and preservation of access in accordance with open science best practices: Software and simulation code will be shared and maintained through GitHub, Sourceforge, and ModelDB with persistent and citable versions of software and code archived in Zenodo (using the GitHub to Zenodo workflow). Olfactory bulb imaging data will be hosted at Senselab’s OdorMapDB. All data generated from experiments and simulations will be archived for up to seven years (or according to local regulations for non-US based teams). In the interest of conducting research according to open science best practices, all data supporting publications and of poten-

tial reuse by the community will be made publicly accessible and archived indefinitely in the CU Scholar repository at the University of Colorado Boulder. CU Scholar is built on the open-source Samvera software, and it has a robust preservation plan that guarantees access to and preservation of data for the lifespan of the repository with post-lifespan contingency plans in place as well. The repository is in the process of obtaining CoreTrustSeal certification, a leading certification for data repository trustworthiness, and is listed in the Re3data registry of research data repositories. The repository provides DataCite DOIs for all data sets, which allow data to be discovered via DataCite’s data set search and API, persistently identified, and cited. CU Scholar provides public access via Globus data transfer to data archived on the PetaLibrary for data sets that are too large to access via direct download. Johnson will assist with preparing data for publication and deposit in CU Scholar and other repositories in accordance with FAIR data principles, and his team will provide ongoing data curation for the data published in the CU Scholar repository. Data published in CU Scholar will provide links to associated publications and archived software and code.

3. Policies

Policies for access and sharing in accordance with open science best practices: All data will be shared upon publication or one year after funding has ended, whichever comes first. Data associated with publications and of potential value for reuse by the community will be made publicly available via the CU Scholar repository, which is openly accessible via HTTP. Large data sets will be publicly discoverable via CU Scholar and openly accessible via Globus data transfer from the PetaLibrary. Data sets in CU Scholar are discoverable via the repository’s search/browse capabilities, indexing in major search engines (e.g., Google), and registries that list CU Scholar and/or its individual data sets (e.g., Re3data registry of research data repositories and DataCite data set search). A directory of the non-digital data and other physical resources generated by the project, including protocols for sharing with team members and external users, will be made available online via the CU Scholar repository.

Policies and provisions for re-use or re-distribution: No restrictions will be placed on the data for re-use or re-distribution other than attribution of the data to the original source. This policy for re-use and redistribution will be included in the human- and machine-readable metadata that accompanies all published data.

4. Use of Vertebrate Animals: Vertebrate animals (mice, *M. musculus*) will be used in each of the IRGs. Mice are an ideal mammalian model system because they are highly olfactory animals and have olfactory circuits that are highly homologous to other vertebrates—including humans—and because genetic-based tools are available and essential to the proposed experiments. Procedures to be performed are provided in the description for each IRG, and involve odorant exposure followed by euthanasia (IRG1); imaging in anesthetized or awake, head-fixed mice (IRGs1 & 3); measurements using chronically-implanted probes in freely-moving mice (IRGs 2 & 3); and conditioning of odor-driven navigation behavior in head-fixed mice (IRG3). The number of animals to be used in each study will be determined based on estimates of the success rates of the various experiments and on statistical power analysis using assumptions of effect sizes, measurement variability and number of data points that can be collected per experiment. This analysis will be done in consultation with biostatisticians at each institution; a typical goal will be achieving a statistical power of 0.8 for comparison of two different populations with an anticipated medium effect size (Cohen’s $d = 0.6$), at a confidence level of $p = 0.05$. In all cases, appropriate protocols will be used to minimize exposure to discomfort, pain or injury, including general anesthetics (isoflurane, ketamine/xylazine), local anesthetics around incision sites (bupivacaine); administration of post-operative analgesics; and monitoring of body weight and signs of stress in conditioned animals. Animals will be euthanized with an overdose of pentobarbital (200 mg/kg), consistent with AVMA Guidelines on Euthanasia of Animals. All procedures will be approved by Institutional Animal Care and Use Committees.